A Goldblatt-Thomason theorem for Compact Polyhedra

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June 30, 2026

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1 Polyhedra

As for the title, we are interested in compact polyhedra. Our claim find its roots in the fundamental interplay between the geometric and the combinatorial nature of said objects, which is perfectly represented by simplicial complexes.

1.1 Polyhedra and maps

Definition 1.1.1. *Simplex*
First consider a set of ***vertices*** $V \subseteq \mathbb{R}^n$ *(for some $n \in \mathbb{N}$).*
We say that

$$\sigma = \left\{ \sum_{i=0}^{k} \alpha_i v_i \mid \alpha_i \in [0,1], \ \sum \alpha_i = 1, v_i \in V \right\}$$

for $v_0, ..., v_k$ vertices in general position (affinely independent) Is called a ***face*** *or* ***simplex*** *with support $v_1, ..., v_k$.*
For a face σ on $v_1, ..., v_k$, a face with support a subset of the vertices is called a ***subface*** *or* ***subsimplex***.
If a face σ has vertices $v_0, ..., v_k$, then we say that $\dim(\sigma) = k$.
We call $\mathrm{RelInt}(\sigma)$ the ***Relative interior*** *of a face σ the interior of the set in the smallest affine subspace of $\mathbb{R}^n$ containing it.*

Definition 1.1.2. *Simplicial Complex*
A ***simplicial complex*** *is defined to be a pair $K = (V, \Sigma)$ where V is a set of vertices and Σ is a set of simplices such that*

- *If $\sigma \in \Sigma$ and σ' is a subsimplex of σ, then $\sigma' \in \Sigma$*
- *The relative interior of any two simplices is disjoint.*

If $K = (V, \Sigma)$ is a simplicial complex we say that $\dim(K) = \max_{\sigma \in \Sigma}(\dim(\sigma))$

This clearly defines a polyhedron in $\mathbb{R}^n$ but in a way that "remembers the faces". We can forget about the structure and define a polyhedron in the following way:

Definition 1.1.3. *Polyhedron*
Let $K = (V, \Sigma)$ be a simplicial complex. Then we call

$$|K| = \bigcup_{\sigma \in \Sigma} \sigma = \bigsqcup_{\sigma \in \Sigma} \mathrm{RelInt}(\sigma)$$

the ***realization*** *of K or the* ***polyhedron*** *associated to K.*
We say that $P \subseteq \mathbb{R}^n$ is a ***polyhedron*** *if there exists a simplicial complex K such that $P = |K|$*

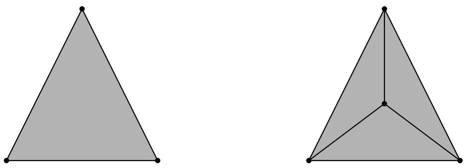
This leads us to two definitions of maps of polyhedra: one that only preserves the information about the simplicial complex, the other informed by the affine construction we did.

Definition 1.1.4. *Simplicial map, PL map*
Let $K = (V, \Sigma), K' = (V', \Sigma')$ be two geometric simplicial complexes.
$f : |K| \to |K'|$ is called a ***Simplicial map*** *(or* ***Polyhedral map****) if*

- *for any point $v \in V$, $f(v)$ is a point of V', and*
- *for any simplex $\sigma \in \Sigma$, $f_{\to}(\sigma)$ is a simplex of Σ'.*

*f is furthermore called PL (****Piecewise linear****) if it is affine when restricted to any face.*

Note that given a simplicial complex, there is a unique polyhedron associated to it, but the other association is not unique:



In the picture above, clearly the two polyhedra are the same (a triangle) but the complex on the left has one face of dimension 2, while the complex on the right has three.

Definition 1.1.5. *Triangulation*
Let P be a polyhedron in $\mathbb{R}^n$, then a simplicial complex K such that $|K| = P$ is called a ***triangulation*** *of P.*

In particular, given a simplicial complex we have a way of generating arbitrarily rich simplicial complexes resulting in the same polyhedron.

Definition 1.1.6. *Barycentric subdivision*
Recall that if $\sigma = \left\{ \sum_{i=0}^{k} \alpha_i v_i \mid \alpha_i \in [0,1],\ \sum \alpha_i = 1, v_i \in V \right\}$, *the point*

$$b_\sigma = \frac{1}{k+1} \sum_{i=0}^{k} v_i$$

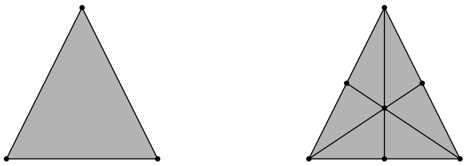
is called the ***barycenter*** *of* σ.
Let $K = (V, \Sigma)$ *be a simplicial complex. Then we call the* ***barycentric subdivision*** *subdivision of* K *the simplicial complex* $K^{(1)}$ *having as vertices the set of* ***barycenters*** *of faces in* Σ

$$V^{(1)} = \left\{ b_\sigma : \sigma \in \Sigma \right\}$$

And as simplices the ones obtained by inductively replacing vertices with the barycenters in descending dimensions.

$$\sigma_j^{(1)} = \left\{ \alpha_1 v_1 + \cdots + \widehat{\alpha_j v_i} + \alpha_j b_\sigma + \cdots + \alpha_k v_k \mid \alpha_i \in [0,1],\ \sum \alpha_i = 1 \right\}$$

Visually,



Theorem 1.1.7. $|K| = |K^{(1)}|$
The geometric realization is invariant under taking barycenters.

Proof. We need to show two inclusions: one is that $\sigma_j^{(1)} \subseteq \sigma$ for all σ and for all j and then, that $\sigma \subseteq \bigcup_j \sigma_j^{(1)}$
For the first claim, suppose $p \in \sigma_j^{(1)}$, without loss of generality $p \in \sigma_1^{(1)}$.
Then

$$\begin{aligned} p &= a_1 b_\sigma + a_2 v_2 + \cdots + a_k v_k \\ &= \frac{a_1}{k+1}(v_1 + \cdots + v_k) + a_2 v_2 + \cdots + a_k v_k \\ &= \frac{a_1}{k+1} v_1 + (a_2 + \frac{a_1}{k+1}) v_2 + \cdots + (a_k + \frac{a_1}{k+1}) v_k \end{aligned}$$

Now

$$\begin{aligned} &\frac{a_1}{k+1} + (a_2 + \frac{a_1}{k+1}) + \cdots + (a_k + \frac{a_1}{k+1}) \\ &= (k+1)\frac{a_1}{k+1} + a_2 + \cdots + a_k \\ &= a_1 + a_2 + \cdots + a_k = 1 \end{aligned}$$

and all of them are positive. This means that any subdivision obtained by replacing a vertex with the barycenter of a face containing it and taking a convex hull is contained in the original face. As the barycentric subdivision is an iteration of this process, $K^{(1)} \subseteq K$
To show the converse, suppose $p \in \sigma$, then

$$p = a_1 v_1 + \cdots + a_k v_k$$

We want to show that there are $b_1, ..., b_k$ such that $p = b_1 v_1 + \cdots + b_j b_\sigma + \cdots + b_k v_k$. If we try to do the substitutions backwards

$$\begin{aligned} &b_j v_j + b_1 v_1 + \cdots + b_k v_k \\ &= \frac{a_j}{k+1} v_j + (a_1 + \frac{a_j}{k+1}) v_1 + \cdots + (a_k + \frac{a_j}{k+1}) v_k \end{aligned}$$

we get a system of equations

$$a_j = (k+1) b_j \quad a_1 = b_1 - b_j \quad \cdots \quad a_k = b_k - b_j$$

To ensure they're all positive we can just pick j as the one realizing the minimum of the b_j's.
Then by induction on the subdivision we may conclude $K \subseteq K^{(1)}$. $\square$

1.2 Posets

Forgetting about the support in $\mathbb{R}^n$ gives us a great combinatorial descryption of simplicial complexes: We have essentially a downward-closed poset.

Definition 1.2.1. *Face Poset*
*Let $K = (V, \Sigma)$ be a simplicial complex. We call $[K] = (\Sigma, \subseteq)$ the **face poset** of K*

$[K]$ is essentially all we need to recover the structure of a simplicial complex up to PL-homeomorphism.

Lemma 1.2.2.
Let K, K' be two simplicial complexes with $[K] \simeq [K']$, then there exists a PL-homeomorphism $K \simeq K'$

Proof. [Col72]Lemma 2.18 □

This means that we can define

Definition 1.2.3. *ASC*
*An **abstract simplicial complex**, or **ASC** is a pair $K = (V, \Sigma)$ where V is a set, and $\Sigma \subseteq \mathcal{P}(V)$ is a collection of subsets of V closed under taking subsets.*
In particular, given a simplicial complex K, we may associate to any face the set of its vertices, resulting in an ASC $\overline{K}$.

Of course we may assign an essentially unique simplicial complex to an ASC.

Lemma 1.2.4.
Let K be an ASC. Then there exists a simplicial complex K' such that $\overline{K'} = K$.

Proof.
We may realize every $d-$simplex to the standard simplex of the correct dimension in $\mathbb{R}^{d+1}$ and glue them according to the intersections. [Col72] has a complete account of it. □

This lemma and the previous one give a correspondence

$$\frac{\{\textbf{Simplicial Complexes}\}}{\text{PL-homeomorphism}} \leftrightarrow \{\textbf{ASCs}\}$$

As to each ASC we may associate a simplicial complex, and any two corresponding ones are PL-homeomorphic.
We want a way to define a(n abstract) simplicial complex from any poset:

Definition 1.2.5. *Nerve*
*Let $(P, \leq)$ be a poset. We call $\mathcal{N}(P)$ the **nerve** of P, the abstract simplicial complex whose vertices are the elements of P, and whose simplices are the flags $p_1 < \cdots < p_k$ of ordered elements of P.*

A fact we want to use freely in the future is

Theorem 1.2.6. *If $K = (V, \Sigma)$ is a simplicial complex, then*

$$\mathcal{N}(\Sigma) = \overline{K^{(1)}}$$

Proof. [Col72] □

1.3 Subdivision theorems

We care about polyhedra as a semantics for modal logic, thus our choice of morphisms is slightly different from the classical framework.
First recall that if we have (binary) kripke frames (W, R), (W', R'), a function $f : W \to W'$ is called a $p-$morphism if and only if it satisfies

1. wRw' implies $f(w)Rf(w')$
2. $f(v)Rw'$ implies that there exists $w \in W$ such that vRw and $f(w) = w'$.

If $R =\leq$ is a partial order, then the two conditions become equivalent to

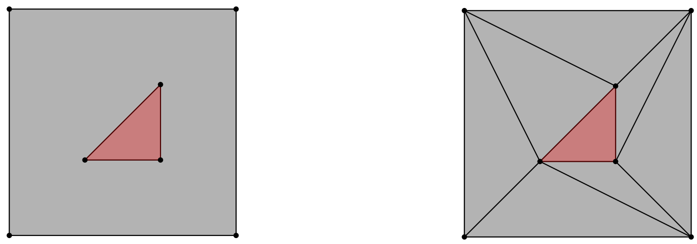
$$\forall w \in W.\ \uparrow f(w) = f(\uparrow w)$$

where $\uparrow w = \{v \in W : w < v\}$.
We refer to this as **upward-closure**.
The proofs of all following lemmas are found in [Dek23].

Lemma 1.3.1. *Subdivisions are p-morphic*
The map $K^{(1)} \twoheadrightarrow K$ sending each face of $\Sigma^{(1)}$ to the (only) face in Σ containing it is a p-morphism.

Lemma 1.3.2. *Common Triangulation*
Let K be a simplicial complex, $K' \subseteq K$ another simplicial complex. Then there exists a subdivision L of K and a subset L' such that L' is a subdivision of K'.

The visualization of this is pretty simple:



Lemma 1.3.3. *Nerves are dense*
Let K be a simplicial complex, L a subdivision of K.
Then there exists $n \in \mathbb{N}$ such that $K^{(n)}$ is (isomorphic to) a subdivision of L.

Lemma 1.3.4. *Subdividing p-morphisms*
Let K, K' be simplicial complexes, $f : K \circ\!\!\twoheadrightarrow K'$ an up-reduction, that is a p-morphism defined on an upward closed (open) subset of K.
Let L' be a subdivision of K', then there exists a subdivision L of K and an up-reduction $L \circ\!\!\twoheadrightarrow L'$ such that

$$\begin{tikzcd} K \arrow[r, "\circ" description, two heads] & K' \\ L \arrow[u] \arrow[r, "\circ" description, two heads] & L' \arrow[u] \end{tikzcd}$$

2 Definability

Here we define the standard versions of our theorem and our semantics, and how we are going to use them.

2.1 Polyhedral semantics

As it is always the case, we have a language, in this case the standard modal language.

Definition 2.1.1. *Modal language Our language is the standard modal language: We assume a (countable) set of propositional variables X and we define inductively*

$$\mathcal{L} = x \in X \mid \bot \mid \varphi \vee \psi \mid \neg\varphi \mid \Diamond \varphi$$

And we interpret our formulas in polyhedra in the following way

Definition 2.1.2. *Polyhedral semantics*
Let P be a (compact) polyhedron. Our ***evaluation*** *is a map $v : \mathcal{L} \to \mathcal{P}\,P$ such that $v(\varphi)$ is the union of relative interior of subpolyhedra of P.*
We say $P, x \vDash \varphi$ if and only if $x \in v(\varphi)$.
Moreover we ask that v satisfies the following properties:

- $v(\bot) = \emptyset$,
- $v(\varphi \vee \psi) = v(\varphi) \cup v(\psi)$,
- $v(\neg\varphi) = P \setminus v(\varphi)$,
- $v(\Diamond \varphi) = \overline{v(\varphi)}$

We say that φ ***is valid in*** *P, written$P \vDash \varphi$ if for any $x \in P$ and for any evaluation v, $P, x \vDash \varphi$*

This allows us to define the **logic** of a polyhedron

Definition 2.1.3.
Let P be a polyhedron. We call the ***Logic*** *of P*

$$\boldsymbol{Log}(P) := \{\varphi \in \mathcal{L} : P \vDash \varphi\}$$

Recall that the kripke semantics for (S4.Grz) modal logic is the one interpreting boolean connectives usually, and interpreting

$$P, p \vDash \Diamond \varphi \iff \exists q \geq p : P, q \vDash \varphi$$

We may connect the two semantics via this theorem

Theorem 2.1.4. *Let P be a polyhedron, $K = (V, \Sigma) \in \mathrm{Triang}(P)$ a simplicial complex realizing to P.*
Then for each $x \in P$ there exists a unique $\sigma_x \in \Sigma$ such that $x \in \mathrm{RelInt}(\sigma_x)$. Then

$$P, x \vDash \varphi \text{ (as polyhedra) if and only if } \Sigma, \sigma_x \vDash \varphi \text{ (as posets)}$$

Moreover

$$\boldsymbol{Log}(P) = \bigcap_{K \in \mathrm{Triang}(P)} \boldsymbol{Log}([K]).$$

Proof.
A complete account may be found in [tGS09] and [Bez+21]. □

2.2 Modally definable classes

The dual question of validity is definability:
Given a set of formulas F, what class of models satisfies it?

Definition 2.2.1. *MD Class*
Let $F \subseteq \mathcal{L}$ be a set of formulas. We call $\boldsymbol{Pol}(F)$ the class of (compact) polyhedra modeling every formula of F, i.e.

$$\boldsymbol{Pol}(F) = \{P \in \boldsymbol{Pol} : \forall \varphi \in F. P \vDash \varphi\}$$

We want to form a Goldblatt-Thomason-Like theorem for this class of spaces.
recall

Theorem 2.2.2. *Goldblatt-Thomason for Finite transitive frames (I)*
A class of finite transitive models C is modally definable if and only if it is closed under

- *Finite disjoint unions*
- *Generated subframes*
- *P-morphic images*

Proof. [BRV01], Chapter 3. □

We will see that 'generated subframes' correspond to open subpolyhedra. As our classes are formed by **compact** polyhedra, then open subpolyhedra wil never be there
We shall use a slightly different but equivalent result:

Corollary 2.2.3. *Goldblatt-Thomason for Finite transitive frames (II)*
A class of finite transitive models C is modally definable if and only if it is closed under

- *Finite disjoint unions*
- *surjective up-reductions*

Where an up-reduction is a partial p-morphism whose domain is a generated subframe.

Proof. Suppose C is closed under (I), $X \in C$ and $f : X \circ\!\!\twoheadrightarrow Y$ an up-reduction Call U the domain of f, then $U \in C$.
$f_{|U} : U \twoheadrightarrow Y$ is a surjective p-morphism, thus Y is p-morphic image of U, thus $Y \in C$.
This means that C is closed under (II).
Conversely suppose that C is closed under (II), let $X \in C$ and and let U be a generated subframe of X. Then the identity of U, seen as a partial map $X \to U$ is an up-reduction. Similarly, if Y is a $p-$morphic image of Y, as X is a generated subframe of X, we may extract an up-reduction. □

Note that this means that the class of models **is always closed under generated subframes** but we may intersect the class with a category that doesn't have them.
We will also make use of the Jankov-Fine axiomatization.

Theorem 2.2.4. *Jankov Fine Let U be a finite rooted frame. Then there exists a formula $\chi(U)$ such that $F \vDash \chi(U)$ if F does **not** up-reduce to U.*

Proof. [BB22] □

Corollary 2.2.5. *A class of finite frames C in a category of models* Mod *can be characterized as*

$$C = \{M \in \mathrm{Mod} : \forall U \notin C \ \textit{rooted}, \ M \vDash \chi(U)\}$$

2.3 Topological connection

Proposition 2.3.1.
The realization $|-|$ enjoys the following properties:

(i) $|U \cup V| = |U| \cup |V|$

(ii) $|U \cap V| = |U| \cap |V|$

(iii) $|U^\circ| = |U|^\circ$

(iv) $|\Sigma \setminus U| = |\Sigma| \setminus |U|$

Proof. They follow immediately from the fact that $|U|$ is a disjoint union. □

Lemma 2.3.2. *up-sets are open*
Let $K = (V, \Sigma)$ be a simplicial complex, $U \subseteq \Sigma$ and consider $|U| = \bigsqcup_{\sigma \in U} \mathrm{RelInt}(\sigma)$
Then $|U|$ is open if and only if U is upward-closed.
Respectively, $|U|$ is closed if and only if U is downward-closed

Proof. We may use the characterization that $|U|$ is closed if and only if $|U| \cup \partial|U| = |U|$
Note that $\partial|U|$ is given by $|U| \setminus \mathrm{RelInt}(|U|) = |U \setminus U^*|$ (where) U^* are the maximal faces. This means that $|U|$ is closed if and only if U contains everything below its maximal faces.
Then $|U|$ is open if and only if $|\Sigma| \setminus |U| = |\Sigma \setminus U|$ is closed thus if $\Sigma \setminus U$ is downward-closed, thus if U is upward-closed. □

3 GTKP

3.1 Face up-reductions

To any poset we can assign to it a topology that is compatible with topological semantics.

Definition 3.1.1. *Alexandrov topology*
Let $(P, \leq)$ be a poset. We call the ***Alexandrov*** *topology on P $\tau \subseteq \mathcal{P}\, P$ where $U \in \tau$ if and only if U is upward-closed.*

We will use it to define a weaker version of the up-reduction between polyhedra.

Definition 3.1.2. *Face up-reduction Let P, Q be (compact) polyhedra and K a triangulation of Q.*
A ***face up-reduction*** *$P \circ\!\!\mapsto Q$ is a partial map $f : P \circ\!\!\twoheadrightarrow [K]$ such that*

(i) The preimage of every open of $[K]$ (in the Alexandrov topology) is an open subpolyhedron of P

(ii) It is ***open***

(iii) It is ***Surjective****.*

Note that

Proposition 3.1.3.
$Q \circ\!\!\twoheadrightarrow [K]$ for any triangulation K.
In particular $Q \circ\!\!\mapsto Q$.

Proof.
Let K be a triangulation of Q and recall that for every point $q \in Q$ there exists a unique face σ_q of K such that $q \in \sigma_q$, thus the map is trivially surjective.
Let f be the map $q \mapsto \sigma_q$.
This means that the face of the polyhedron $|\sigma| = f^{\leftarrow}(\sigma)$.
If U is an upward-closed set of $[K]$, then $[K] \setminus U$ is downward-closed, thus is a compact polyhedron. This means that $f^{\leftarrow}([K] \setminus U) = Q \setminus f^{\leftarrow}(U)$ is compact, thus closed, therefore $f^{\leftarrow}(U)$ is open and is the union of faces, thus an open subpolyhedron.
This same argument shows that the map is open: If $P \subseteq Q$ is an open subpolyhedron, then $Q \setminus P$ is a compact polyhedron, thus $[K] \setminus f(P)$ is downward-closed, thus $f(P)$ is upward-closed, i.e. open. ☐

Corollary 3.1.4.
If there exists an up-reduction, i.e. an open partial simplicial map $P \circ\!\!\twoheadrightarrow Q$ whose domain is an open subpolyhedron of P then $P \circ\!\!\mapsto Q$

We may also make explicit the following fact:

Lemma 3.1.5.
Let $f : P \circ\!\!\mapsto Q$ be a face up-reduction, $U \subseteq^o P$ is the domain of f and L is the triangulation of Q such that $f : U \circ\!\!\twoheadrightarrow [L]$.
Then there exists a triangulation K of P such that

- *U is the union of relative interiors of faces of K, i.e. $U = \bigcup_{\sigma \in K_U} \mathrm{RelInt}\, \sigma$.*
- *$\tilde{f} : [K] \circ\!\!\twoheadrightarrow [L]$ defined as $\tilde{f}(\sigma_p) = f(p)$ is an up-reduction of posets.*

Proof.
Note that $[L]$ is a **finite** poset and that $\{f^{\leftarrow}(l) : l \in L\}$ is a collection of possibly open polyhedra in P. By the common triangulation lemma, we may extract a common triangulation K of all of them.
Now, recall that for each point $p \in P$ there exists a unique face σ_p such that $p \in \mathrm{RelInt}(\sigma_p)$. This allows us to define $\tilde{f}$ as above. Uniqueness gives us that $\tilde{f}$ is well-defined, and $f(\uparrow \sigma_p) = \uparrow f(p)$ by openness of f. ☐

Lemma 3.1.6 (Up-reductions refute formulas)**.** *Let P be a polyhedron, S a poset, $P \circ\!\!\twoheadrightarrow S$ an up-reduction.*
If $S \nvDash \varphi$ then $P \nvDash \varphi$ as well.

Proof. The proof goes essentially as the topological case in [Gab01] ☐

3.2 Goldblatt-Thomason for Compact Polyhedra

Theorem 3.2.1. *GTKP*
A class of compact polyhedra C is modally definable if and only if it's closed under finite disjoint unions and face up-reductions, i.e.

(I) if $P, Q \in C$, then $P \sqcup Q \in C$

(II) if $P \in C$ and $P \circ\!\!\mapsto X$, then $X \in C$.

Proof ($\Longrightarrow$).
Let C be a modally definable class, i.e. $C = \{P \in \mathbf{Pol} : P \vDash F\}$ for some set of formulas F.
Closure under disjoint union is trivial, if $P, Q \in C$, then $P \sqcup Q \vDash F$.
If $P \in C$ and $P \circ\!\!\mapsto X$, then suppose $X \notin C$. By definition this means that there exists a formula $\varphi \in F$ such that $X \nvDash \varphi$.
Spelling out the validity condition, there exists an evaluation $\nu : \mathcal{L} \to \mathcal{P}X$ and a point $x \in X$ such that $X, x \nvDash \varphi$.
By the simultaneous triangulation lemma, we may select a triangulation S of X such that $\nu(\varphi)$ is triangulated by S.
Now call $K = (V, \Sigma)$ the triangulation of X realizing the face up-reduction $P \circ\!\!\mapsto X$.
By the 'nerves are dense' lemma, we know that there exists a barycentric subdivision $K^{(n)}$ of K subdividing S.
We also know that we can extend any up-p-morphism $P \circ\!\!\twoheadrightarrow \Sigma$ to an up-p-morphism $P \circ\!\!\twoheadrightarrow \Sigma^{(k)}$ for any $k \in \mathbb{N}$.
This means that we have an up-p-morphism $P \circ\!\!\twoheadrightarrow \Sigma^{(k)} \circ\!\!\twoheadrightarrow S$, therefore $P \nvDash \varphi$, leading to a contradiction. □

Proof ($\Longleftarrow$).
Let C be a class of polyhedra closed under (I) and (II), call $L := \mathbf{Log}(C)$ and call $C' = \mathbf{Pol}(L)$.
Clearly $C \subseteq C'$. Let $X \in C'$, we want to show that $X \in C$.
We know that $X \vDash L$, and let $K = (V, \Sigma) \in \mathrm{Triang}(X)$.
Call v_i the vertices of Σ. We know that

$$\bigsqcup_i (\uparrow v_i) \twoheadrightarrow \Sigma$$

is a p-morphism.
Consider $\chi_i := \chi(\uparrow v_i)$ the Jankov-Fine formula for the (rooted) subposet.
As $(\uparrow v_i) \nvDash \chi_i$, we know that $\Sigma \nvDash \chi_i$, as all $(\uparrow v_i)$ are generated subframes of Σ.
In particular $X \circ\!\!\twoheadrightarrow \Sigma$, thus $X \nvDash \chi_i$, for all i.
This means that none of the χ_i are elements of L.
This means that for each χ_i there exists some $P_i \in C$ such that $P_i \nvDash \chi_i$, thus $P_i \circ\!\!\twoheadrightarrow (\uparrow v_i)$.
This means that $P := \bigsqcup_i P_i$ admits an up-p-morphism to $\bigsqcup_i (\uparrow v_i)$, and we can take the composition with $\bigsqcup_i (\uparrow v_i) \twoheadrightarrow \Sigma$ to find an up-reduction $P \circ\!\!\twoheadrightarrow \Sigma$, thus a face up-reduction $P \circ\!\!\mapsto X$.
By (I) we know $P \in C$, and by (II) $X \in C$. □

Corollary 3.2.2. *A class of polyhedra closed under finite disjoint unions and simplicial up-reductions is modally definable, where a simplicial up-reduction is an open simplicial partial map $f : P \circ\!\!\twoheadrightarrow Q$ whose domain is an open subpolyhedron of P.*

Proof. Follows from Corollary 3.1.4: If $f : P \circ\!\!\twoheadrightarrow Q$ is a simplicial up-reduction, then there exists a face up-reduction $P \circ\!\!\mapsto Q$ by composing with the face p-morphism onto any face poset. □

The theorem above is strengthened by the following lemma

Lemma 3.2.3 (Any Triangulation)**.**
Let $P \circ\!\!\twoheadrightarrow \Sigma$ for a triangulation $L = (V, \Sigma)$ of Q, and let $L' = (V', \Sigma')$ be a different triangulation of Q. Then there exists an up-reduction $P \circ\!\!\twoheadrightarrow \Sigma'$.

Proof. Let $f : P \circ\!\!\twoheadrightarrow \Sigma$ be an up-reduction with domain $U \subseteq P$.
Then we can extend f to $f^{(1)} : P \circ\!\!\twoheadrightarrow \Sigma^{(1)}$ in the following way:
Take the triangulation K of P realizing f as a simplicial map $\tilde{f}$ (as per lemma 3.1.5)
Then utilize the subdivision lemma (lemma 1.3.4) to extract an up-reduction $K' \circ\!\!\twoheadrightarrow L^{(1)}$.
Now, utilizing lemma (1.3.3) we may repeat this n times, such that $L^{(n)}$ is a subdivision of L'.
This means that we have a map $h : K'^{\cdots\prime} \circ\!\!\twoheadrightarrow L^{(n)} \twoheadrightarrow L'$
As $K'^{\cdots\prime}$ is a triangulation of P we can extract an up-reduction $g : P \circ\!\!\twoheadrightarrow L'$ by setting it as locally constant on the relative interior of the faces of P. □

In other words, if a face up-reduction exists to a triangulation then one exists to any triangulation, even with the same domain.

4 Open questions

4.1 A refinement of GTKP

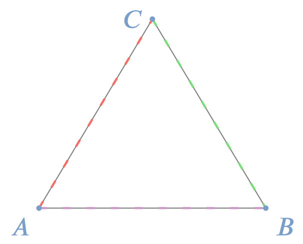
The result we have spent the most time trying to obtain is the following refinement

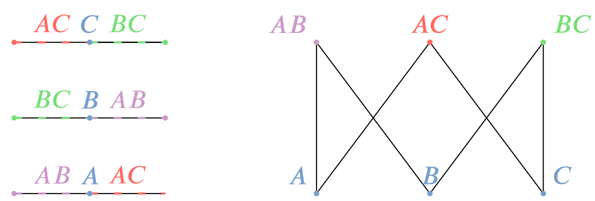
Conjecture 4.1.1. *A class of compact polyhedra is modally definable if and only if it is closed under finite disjoint union and simplicial up-reductions.*

The procedure we have in mind to prove the equivalence between the current result and the one above is the following:

- Consider the face up-reduction $f : P \mathrel{\circ\!\!\rightarrowtail} Q$ realized as $f : P \mathrel{\circ\!\!\twoheadrightarrow} [L]$;
- Realize f as a map $\tilde{f} : [K] \mathrel{\circ\!\!\twoheadrightarrow} [L]$;
- Construct an up-reduction $\tilde{g} : K \mathrel{\circ\!\!\twoheadrightarrow} L$ "informed by $\tilde{f}$" that is an up-reduction and a map of simplicial complexes;
- Realize $\tilde{g}$ as a simplicial up-reduction $g : P \mathrel{\circ\!\!\twoheadrightarrow} Q$.

If $\tilde{f}$ were (or densely contained) a map of (abstract) simplicial complexes and not only one of posets our work would be done: It is possible to realize any map of ASCs into a simplicial map of their polyhedra. In general though this is not the case: Consider P to be three segments, and Q to be the triangle (boundary of the 2−simplex), and consider

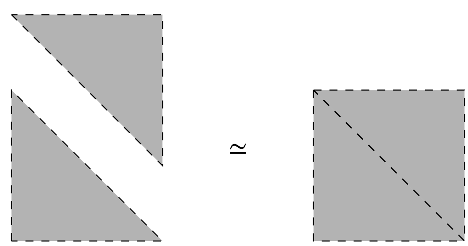




The hasse diagram (bottom right) is L, the three lines are P and the triangle is Q. The map is indicated by the coloring.
It can be checked that it is a p-morphism but clearly it's not a map of simplicial complexes as the image of vertices are not vertices themselves.

4.2 The precompact case

How do things change if we allow for open polyhedra? We gain access to generated subframes, but then the relationship between polyhedra and simplicial complexes becomes more loose:
Consider the poset made of one point, it is the poset of the maximal face of any simplex.
Moreover, even trying to circumvent this problem by considering their closure, indeed a simplicial complex, we can encounter the case where the combinatorial description is not PL-invariant.
For example consider the disjoint union of two triangles.



The two objects are PL-homeomorphic but their simplicial complexes are not.

4.3 The infinite case

An interesting question is what happens when we allow for arbitrarily sized simplicial complexes, both in dimension and number of simplices. Then the class is not of transitive finite frames but arbitrary ones.

- What is the geometric realization of an arbitrarily big abstract simplicial complex?
- What is the ultrafilter extension of an arbitrarily big simplicial complex?
- How do we describe such ultrafilter extension geometrically?
- (abstract) simplicial complexes allow for a coalgebraic description: They are a class of $\mathcal{P} \circ \mathcal{P}$ coalgebras over sets (the map $V \to \mathcal{P}\mathcal{P}V$ is the one associating to every point the up-set of simplices containing it). Do completeness and definability results coincide to the usual ones in that setting?

5 Applications

5.1 Some Modally definable classes

5.2 Some Not modally definable classes

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